

Case Study 1

Sample Report

Data is from two different sources, data of daily site configurations collected by ABC and visitor duration information collected by an external website tracking company. The variables in the ABC data include the date, an indicator for if a news article was posted, the number of advertisements on the web page, the number of links to predictive modeling articles, and the topic of the first article linked to on the page. The tracking data includes the date of a visitor, the duration of the visit, an indicator for if the visit was a first time or not, and the sex of the individual. The ABC data was collected for every day in 2017, whereas the tracking data only includes about 5 months of 2017.

The goal is to build a predictive model for the average daily duration of new visitors to the website. This target variable was not available in the given form of the data. To find these values, the tracking data was filtered to only records that indicated new visitors, and then the average duration was calculated by day. This will become the target variable in the predictive model we are building. The target variable, average duration of new visitors by day, is plotted in Figure 1.

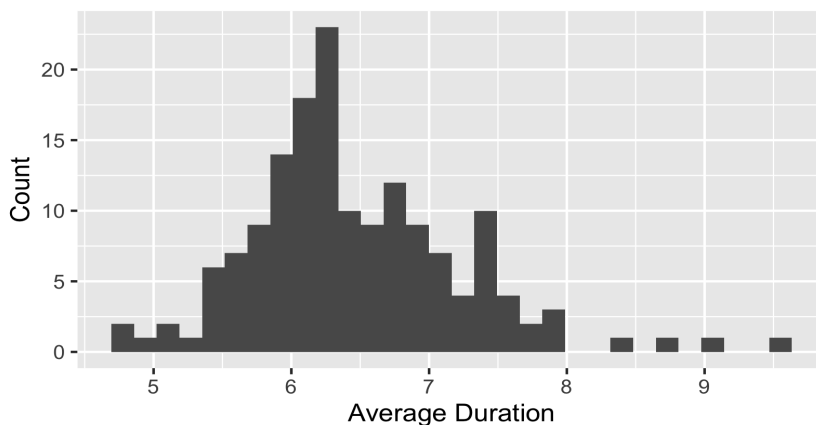


Figure 1. Histogram of Average Duration of Site Visit

Several steps were taken to ensure the data was ready for building a model. Because the tracking data was only five months out of the year, the two data sources were combined, but only keeping the records that matched dates between the two. Because we can only use records with a target variable present to train a predictive model, it makes sense to combine the data this way.

The daily site configuration data was recorded in two different chunks, being

split at the middle of the year. In the first half of the year, the headliner topic, the topic of the top article link, was recorded differently than in the second half of the year. The topics that are meant to be the same were matched up. For example, “machine learning” in the first half is the same as “ML” in the second half, so “ML” was recoded to match “machine learning.” This was repeated for other values. The specific way this variable is recoded is not important, but it is important to make sure there are consistent factor levels so that the model is able to discern the effect of the specific factor levels.

There were several instances of the number of advertisements being well outside the range of the rest of the data. While most advertisement counts were between 3 and 13, four of them had values of 70. It was explained that this was experimental, but would not happen again. It is best to build a predictive model using data that is representative of the new data that we will see in the future. Because this scenario is not going to be repeated, it was decided that these records with 70 advertisements should be removed.

Some records had missing values when marking when a news article was shared on the webpage that day. While these records could possibly be thrown out, the remainder of the records for those days was still useful, so instead it was decided to impute the missing values, meaning that values were given to each missing value using a method that will make good estimates of what those values would have been. In this case, the predictor variables for the rest of the complete records were used to build a simple [Python:logistic regression]

[R:regression tree] model with *news* as the target variable. Finally, the day of the week was extracted from the date values that were provided in the data.

One very powerful predictor could have been the number of individuals who come to a website in a given day. This would not be appropriate to include in a predictive model because the purpose of a predictive model is to make predictions for events that have not yet occurred. The number of people who visit a site in a given day would not be available when the predictive model is being used to predict, and therefore the model should not be built using the number of visitors. This variable was deleted.

The information for the sex of each individual is available in the tracking data. This data is not available at the daily scale where several of each category visited the site each day. That information could be used in a few ways. The percentage of daily visitors of each sex could be used as a predictor in the model. Also, the data could be split up entirely and separate models could be fit for visitors who are specifically male or female. The purpose of this type of model could be to explicitly attract a certain demographic. Perhaps ABC wishes to find ways to have more female users staying longer when visiting their website. Then a model could be built to determine which factors are positively associated with the daily average duration of female visitors. Using sex in this case would be appropriate. Ignoring the sex variable would also be appropriate. There are no pricing, benefits, or placement decisions based on this model, and the only individual specific information used in the model is averaged over a group of individuals, so there is likely no hidden biases in the model that will negatively affect a particular sex. In this case we decided to not use the sex variable in the final data set.

Tables 1-3 provide a summary of the variables in the final data set.

Table 1

	date	news	num_ads	num_links	num_visits
Min.	2017-03-15 00:00:00	0.0000	3.000	6.000	37.00
1st Qu.	2017-04-26 00:00:00	0.0000	6.000	8.000	51.00
Median	2017-06-03 00:00:00	1.0000	8.000	8.000	69.00
Mean	2017-06-02 14:35:17	0.7386	8.261	8.366	71.63
3rd Qu.	2017-07-11 00:00:00	1.0000	11.000	9.000	91.00
Max.	2017-08-18 00:00:00	1.0000	13.000	10.000	117.00

Table 2

day_of_week	
Friday	22
Monday	22
Saturday	21
Sunday	21
Thursday	22
Tuesday	22
Wednesday	23

Table 3

headliner_topic	
machine learning	32
probability	9
regression	43
technology	27
visualization	42